

```

import pandas as pd
from transformers import pipeline
reviews_data = [
# Bollywood
{'movie': '3 Idiots', 'source': 'Bollywood',
'review': 'An absolute masterpiece! Makes you laugh and cry. '
'Aamir Khan delivers one of his finest performances ever.'},
{'movie': '3 Idiots', 'source': 'Bollywood',
'review': 'Overrated and preachy. The message was fine but '
'execution felt too theatrical and unrealistic.'},
{'movie': 'Dangal', 'source': 'Bollywood',
'review': 'Incredibly inspiring! Geeta and Babita brought tears '
'to my eyes. Best sports film ever made in India.'},
{'movie': 'Dangal', 'source': 'Bollywood',
'review': 'Good film but very long. Some scenes dragged unnecessarily.'},
# Hollywood
{'movie': 'Inception', 'source': 'Hollywood',
'review': 'Mind-bending and visually stunning. Nolan at his absolutebest.'},
{'movie': 'Inception', 'source': 'Hollywood',
'review': 'Too complicated. I had no idea what was happening half thetime.'},
{'movie': 'Parasite', 'source': 'Hollywood',
'review': 'Perfect film. The class commentary is brilliant. Deserved everyOscar.'},
{'movie': 'Parasite', 'source': 'Hollywood',
'review': 'I expected more from an Oscar winner. Ending felt rushed.'},
]
df = pd.DataFrame(reviews_data)

```

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# Load once – not inside a loop (that would reload weights every iteration)
sentiment = pipeline('sentiment-analysis',
model='distilbert-base-uncased-finetuned-sst-2-english')
# Batch all reviews at once – faster than one-by-one
results = sentiment(df['review'].tolist())
df['sentiment'] = [r['label'] for r in results] # POSITIVE / NEGATIVE
df['confidence'] = [r['score'] for r in results] # 0.0 to 1.0
# sentiment_score: 1.0 = very positive, 0.0 = very negative
df['sentiment_score'] = df['confidence'].where(
df['sentiment'] == 'POSITIVE',
1 - df['confidence']
)
print(df[['movie', 'sentiment', 'confidence']].to_string(index=False))

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/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable huggingface_hub.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub.
config.json: 100% 629/629 [00:00<00:00, 53.4kB/s]

model.safetensors: 100% 268M/268M [00:02<00:00, 153MB/s]

Loading weights: 100% 104/104 [00:00<00:00, 3332.63it/s]

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 2.67kB/s]

vocab.txt: 100% 232k/232k [00:00<00:00, 7.02MB/s]

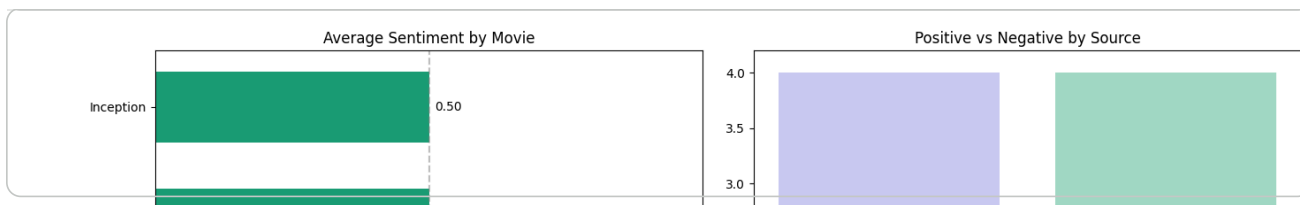
   movie sentiment  confidence
3 Idiots  POSITIVE    0.999879
3 Idiots  NEGATIVE    0.999016
Dangal    POSITIVE    0.999878
Dangal    NEGATIVE    0.992343
Inception POSITIVE    0.999849
Inception NEGATIVE    0.999590
Parasite  POSITIVE    0.999857
Parasite  NEGATIVE    0.998913

```

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import matplotlib.pyplot as plt
fig, axes = plt.subplots(1, 2, figsize=(14, 6))
# Chart 1: Average sentiment score per movie
ax1 = axes[0]
movie_scores = df.groupby('movie')['sentiment_score'].mean().sort_values(ascending=False)
colours = ['#1D9E75' if v >= 0.5 else '#D85A30' for v in movie_scores.values]
bars = ax1.barh(movie_scores.index, movie_scores.values, color=colours, height=0.6)
ax1.set_xlim(0, 1)
ax1.axvline(0.5, color='gray', linestyle='--', alpha=0.5)
ax1.set_title('Average Sentiment by Movie')
for bar, val in zip(bars, movie_scores.values):
    ax1.text(val+0.01, bar.get_y()+bar.get_height()/2, f'{val:.2f}', va='center')
# Chart 2: Positive vs Negative split by source
ax2 = axes[1]
for source, colour in [('Bollywood', '#7F77DD'), ('Hollywood', '#1D9E75')]:
    subset = df[df['source'] == source]
    pos = (subset['sentiment'] == 'POSITIVE').sum()
    neg = (subset['sentiment'] == 'NEGATIVE').sum()
    ax2.bar(source, pos, color=colour, alpha=0.9)
    ax2.bar(source, neg, bottom=pos, color=colour, alpha=0.4)
    ax2.set_title('Positive vs Negative by Source')
plt.tight_layout()
plt.savefig('sentiment_analysis.png', dpi=150, bbox_inches='tight')
plt.show()

```



```
def analyse_review(review_text, movie_name='Unknown'):  
    result = sentiment(review_text)[0]  
    label = result['label']  
    conf = result['score']  
    if label == 'POSITIVE':  
        score = conf  
        verdict = 'LOVED IT' if conf > 0.95 else 'LIKED IT'  
    else:  
        score = 1 - conf  
        verdict = 'HATED IT' if conf > 0.95 else 'DISLIKED IT'  
    print(f'Movie : {movie_name}')    print(f'Verdict : {verdict}')    print(f'Score : {score:.1%} positive')    print(f'Review : {review_text[:80]}...')    return {'movie': movie_name, 'verdict': verdict, 'score': score}  
analyse_review(  
    'Rajkummar Rao was phenomenal. A raw, honest performance. '  
    'Easily the best Hindi film of the decade.',  
    'Stree'  
)
```

```
Movie : Stree  
Verdict : LOVED IT  
Score : 100.0% positive
```